V O L T S L O T

VoltSlot

EV Charging Station Management and Slot Reservation System

Automated Test Report

Student name	[Enter student name]
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Academic year	[Enter academic year]

Prepared from the VoltSlot source code · 2026-10-14

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Execution Summary

Results from commands executed for this documentation task; these are not copied from an earlier report.

Execution date: 2026-10-14. Backend test suite completed before frontend tests. Frontend test command ran Vitest and Playwright.

Suite / check	Command	Observed result
Backend Node test suite	`cd backend; npm test`	**PASS** 71 tests; 18 suites; 71 passed, 0 failed, 0 skipped; exit code 0.
Frontend unit/component suite	`cd frontend; npm test` (Vitest phase)	**PASS** 31 tests in one file; 31 passed; exit code 0.
Frontend browser E2E suite	`cd frontend; npm test` (Playwright phase)	**PASS** 8 tests passed, 0 failed; exit code 0.
Production build	`cd frontend; npm run build`	**PASS** Vite production build was executed for report verification.
Manual UI test / real database test	Not run	Not claimed. No live MongoDB data was changed.

The first Playwright attempt had 7 passes and 1 failure: a wrong-password response appeared as an expired-session message. The login page was corrected to display the invalid-credentials response, and the full frontend command was rerun successfully (31 unit tests and 8 browser tests passed).

Verified Test Cases

Area	Coverage represented in source tests	Observed result
Authentication	Valid/invalid login, customer registration, role assignment rejection, JWT errors, protected routes, profile updates, rate limits.	Backend tests passed; frontend registration/login and role flows passed in Vitest/E2E.
Stations	Public listing filters and availability, validation, admin create/update/delete, dependent-record deletion protection.	Backend tests passed; admin browser flow passed with API stubs.
Slots	Public ordered slot listing, admin authorization, create/update/status changes, duplicate schedule, invalid IDs, dependency protection.	Backend tests passed; admin Playwright management flow passed with API stubs.
Bookings	Customer-only reservation, unavailable/invalid inputs, future/active station checks, repeated/concurrent same-slot conflict, list scope, ownership, cancellation, completion.	Backend tests passed using isolated mocked Mongoose operations. Distinct overlapping slot records are not covered.

Area	Coverage represented in source tests	Observed result
Analytics	Admin dashboard aggregate data and authorization; customer dashboard own-data scope and authorization.	Backend tests passed; frontend dashboards passed with stubbed responses.
Security middleware	CORS policy, security headers, oversized request body behavior.	Backend tests passed.
User reservation experience	Customer session selects available future slot, submits energy amount, sees confirmation; customer browsing and booking history flow.	Frontend Vitest and customer Playwright flow passed with API stubs.

Test Boundaries and Unresolved Failures

- Backend route/model tests are automated but use mocked Mongoose/database methods. They do not demonstrate behavior against a real Atlas instance, transaction isolation, or a deployed MongoDB configuration.
- Playwright E2E tests start the Vite frontend and intercept API calls with fixtures. They demonstrate rendered browser flows and client behavior but do not verify persistence or actual API network integration.
- Vitest uses mocked client modules where appropriate; these are component/frontend tests.
- No manual browser test or live database CRUD test was run for this report. No screenshots are claimed.
- All test commands listed above completed successfully in this run; there are no unresolved failures from that run.
- The tests cover concurrent attempts for the same slot ID; they do not cover distinct slot records with partially overlapping time ranges. The implementation has no station-level overlap exclusion.
- No real MongoDB records were read or modified during documentation generation.

Reproduction Commands

```
cd backend
npm test

cd ../frontend
npm test
npm run build
```

Evidence: terminal results from this documentation task. The generated report distinguishes automated tests from unrun manual/live-data validation.